

Data-Driven Investment Risk Prediction Using Machine Learning and SHAP-Based Interpretability

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Investment risk assessment is a crucial part of financial decision making and can be done manually based on financial analysis and ratios. However, such methods of risk assessment may be very subjective and hard to implement on a large scale. This study provides an alternative approach based on applying machine learning techniques to financial indicators provided by a firm.

In order to build multiple machine learning models (Logistic Regression, Decision Tree, Random Forest), financial indicators of companies from a secondary database have been used. Out of these models, Random Forest has proven its efficiency since it gave about 95-97% of accuracy in assessing investment risk.

In order to provide explainability for this model, SHAP (SHapley Additive exPlanations) approach has been used for analyzing feature importance for each predicted risk level. Moreover, a real-time risk scoring system has been developed which allows assigning values to prediction probabilities in a range from 0 to 100.

The presented model is able to assess investment risk effectively, provide explanations for results, and classify risks into low-, medium-, and high-risk levels. (Abstract)

Keywords— Machine Learning, Investment Risk Assessment, Random Forest, Financial Data Analysis, SHAP, Explainable AI, Risk Scoring

I. INTRODUCTION (HEADING 1)

Decision-making for investment inevitably requires uncertainty, hence the importance of risk assessment in any financial analysis. Traditional approaches have been based on financial ratios analysis and human expertise, both of which are subjective and cannot be scaled in large data settings.

The rise of data analytics and machine learning enables automation in the process of risk assessment. Machine learning algorithms are able to detect complex non-linear correlations among financial variables, leading to better accuracy than traditional approaches.

In this research, the objective is to create a machine learning framework for assessing investment risks by predicting risk classes based on financial variables. Apart from being a classifier, the proposed model utilizes explainable AI concepts, ensuring interpretability and reliability of the output results.

The study also proposes a risk scoring scheme that converts the classification outcome into a score value.

II. LITERATURE REVIEW

1. Initial researches into financial risk prediction showed that statistical modeling was extensively used for bankruptcy prediction, where Logistic Regression was one of the models applied to provide interpretability of findings, yet being ineffective at accounting for non-linear dependencies between variables in the dataset [1].
2. Ensemble learning, especially Random Forest, gained popularity among researchers and practitioners when dealing with financial risk assessment because of its capacity to work with high-dimensional data and increasing predictivity of results [1].
3. Machine learning research into financial distress prediction provided the evidence for the effectiveness of utilizing advanced algorithms that helped to recognize patterns in financial indicators of risks and outperform traditional models [3].
4. Research on machine learning-based bankruptcy prediction stressed the importance of financial indicators such as profitability, liquidity, and leverage in evaluating the risks associated with companies and significantly increased the predictive power [4].
5. Finally, the emergence of Explainable Artificial Intelligence (XAI) enabled the introduction of new approaches for interpreting black box models, including using the Shapley Additive Explanations (SHAP) method [2].

III. PROPOSED SYSTEM

1. A. System Architecture

2. As illustrated in Figure 1, the system architecture comprises different layers:
3. Input Layer: Input is the financial data of companies, which includes several financial metrics like profitability, liquidity, and leverage ratios.
4. Preprocessing Layer: Preprocessing of the financial data is performed, including data cleansing, normalization, and feature scaling.
5. Feature Processing Layer: Features are processed by arranging the financial metrics in a matrix form. Feature selection and feature scaling are used to prepare the features for training the machine learning algorithms.
6. Model Layer: Several machine learning models, including logistic regression, decision tree, and random forest algorithms, are utilized. Model training can be done offline and real-time predictions can be made.
7. Prediction Layer: The output from the machine learning model is the probability of high-risk and low-risk categories for a company.
8. Explainability Layer: SHAP is used to understand the importance of each feature that contributes to the prediction made by the algorithm.
9. Output Layer: Outputs from the system include:
 10. Risk Category (Low/Medium/High)
 11. Risk Score (0-100)
 12. Explanation of Features

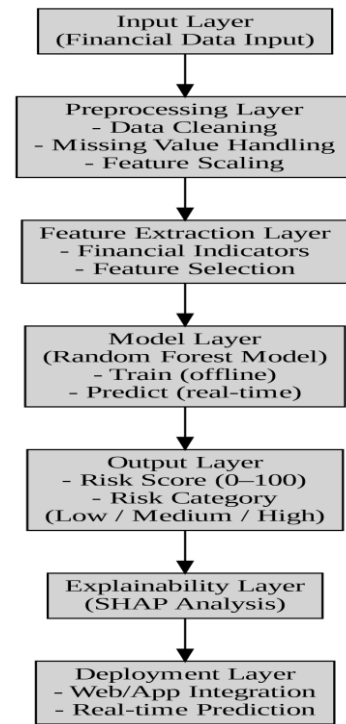


Fig. 1 System Architecture

B. Workflow

13. As shown in Figure 2 below, the flow will be:

14. Data Input
15. Data Preprocessing
16. Feature Engineering
17. Training/Prediction
18. Risk Probability Calculation
19. Risk Score Computation
20. SHAP Explanation
21. Output

C. Innovative Nature of the System

22. Some of the features that will make the proposed system innovative include:
23. The application of machine learning with Explainable AI
24. Transformation of classification output into a continuous risk score
25. Real-time prediction

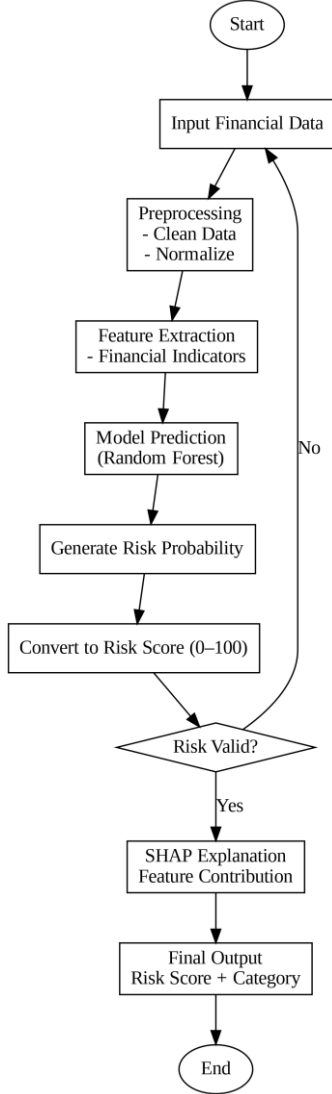


Fig. 2 System Flowchart

IV. RESULTS

A. Model Performance

The performance of the suggested investment risk evaluation system was analyzed based on three algorithms, namely Logistic Regression, Decision Tree, and Random Forest. The analysis was performed by calculating the traditional classification parameters such as accuracy, precision, recall, and F1-score.

The dataset used for this experiment is imbalanced since there are many more samples of low risk (Class 0: 1313 samples) than high risk (Class 2: 51 samples).

The experimental results can be found in Table I.

TABLE I. MODEL PERFORMANCE

Model	Accuracy
Logistic Regression	96.40%
Decision Tree	94.79%
Random Forest	96.63%

From the figure below (Fig. 3), it can be seen that Random Forest performed well in terms of having high accuracy compared to the other two algorithms used, which shows how well it performs in terms of detecting non-linear relationships in the financial dataset.

	Model	Accuracy
0	Logistic Regression	0.964076
1	Decision Tree	0.953079
2	Random Forest	0.967009

Fig. 3 Accuracy Comparison of Models

B. Classification Performance Analysis

A thorough assessment of the model performance in terms of classes provides essential information about the efficiency of each classifier.

The Logistic Regression model showed high precision and recall metrics of 0.97 and 0.99, respectively, for the majority class. However, it showed poor performance concerning the minority class, where the recall score was only 0.20, which means that the algorithm prefers low-risk companies.

The Decision Tree model exhibited more balanced behavior than Logistic Regression by showing a recall of 0.33 for the minority class. However, it had lower accuracy, which means that its generalization capabilities were not sufficient.

The Random Forest model provided the highest overall metrics with an accuracy of 96.63%. As can be seen from Fig. 4, the model showed the highest recall metric for the majority class of 1.00, providing almost perfect discrimination of low-risk companies. However, the recall rate was still low for the minority class of 0.14.

Logistic Regression					
Accuracy: 0.9640762463343109					
	precision	recall	f1-score	support	
0	0.97	0.99	0.98	1313	
2	0.56	0.20	0.29	51	
accuracy			0.96	1364	
macro avg	0.76	0.59	0.64	1364	
weighted avg	0.95	0.96	0.96	1364	

Decision Tree					
Accuracy: 0.9479472140762464					
	precision	recall	f1-score	support	
0	0.97	0.97	0.97	1313	
2	0.31	0.33	0.32	51	
accuracy			0.95	1364	
macro avg	0.64	0.65	0.65	1364	
weighted avg	0.95	0.95	0.95	1364	

Random Forest					
Accuracy: 0.966275659824047					
	precision	recall	f1-score	support	
0	0.97	1.00	0.98	1313	
2	0.78	0.14	0.23	51	
accuracy			0.97	1364	
macro avg	0.87	0.57	0.61	1364	
weighted avg	0.96	0.97	0.95	1364	

Fig. 4 Model Comparison

C. Feature Importance and SHAP Analysis

Interpretation of the model's results was done by applying the SHAP analysis (SHapley Additive exPlanations), where feature importance in specific predictions was identified.

Fig. 5 SHAP Analysis

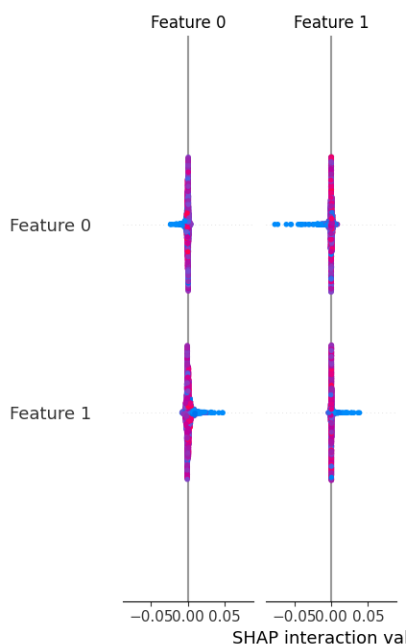


Fig. 5 SHAP Evaluation

According to the results shown in Fig. 5, features like income-expense ratio and non-industrial income/revenue ratio have negative effects on risk prediction since they represent profit levels that decrease risk.

On the other hand, factors like current liabilities to equity, borrowing dependence, and debt-to-equity ratio have a positive contribution to risk, which means that the more

financial obligations a firm has, the more probable it will be labeled as a high-risk firm.

These findings conform to financial theories, which state that high financial leverage and low financial stability lead to high investment risks. The SHAP plot also proves that the model learns valuable financial patterns and not random data associations.

D. Evaluation of the Risk Scoring System

The created risk scoring system uses the probability values produced by the model to assign risk scores (0–100), thus facilitating:

- Risk assessment at any time
- Classification into:
- Low-Risk Category
- Medium-Risk Category
- High-Risk Category

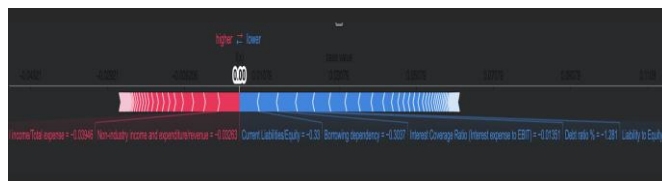


Fig. 6 Risk Scoring

E. Feature Importance Analysis

Significance of financial attributes as an investment risk predictor was assessed using the Random Forest algorithm.

As depicted in Fig. 7, “Net income to stockholders’ equity” is the attribute with the greatest influence, which demonstrates that profitability dominates as a predictor of investment risk. The higher the value of this attribute, the less likely financial risk is because profitable companies are perceived as financially sound organizations.

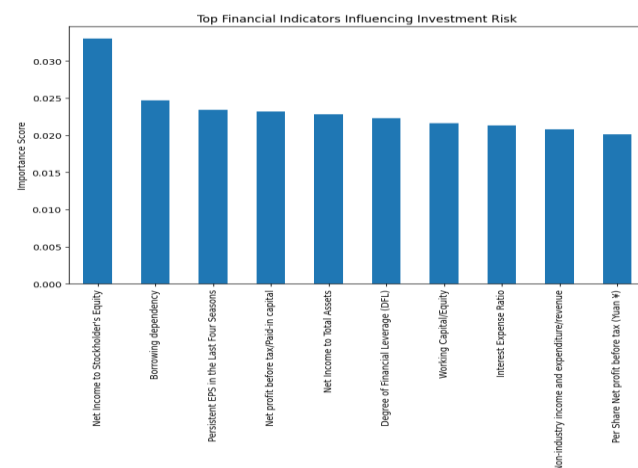


Fig. 7 Top Financial Indicators Influencing Investment Risk

Borrowing dependency and EPS persistency for the past four seasons also have an important contribution to the model prediction (as shown in Fig. 7). This implies that not only is

the financial stability factor important but also the persistent earnings.

Moreover, the financial ratios such as net profit before tax/paid-in capital, net income/total assets, and degree of financial leverage (DFL) also contribute greatly to the model.

In summary, Fig. 7 indicates that the model considers profitability, leverage, and EPS persistence as the most important aspects in investment risk prediction.

F. Comparison of Financial Indicators

The comparison between selected financial indicators was done to determine the consistency of the identified features' importance for different models.

As presented in Fig. 8, there is a consistent ranking for the selected financial indicators where Net Income/Stockholder's Equity still has the highest importance score. Thus, this proves that the model can correctly determine the financial indicator importance consistently.

Also, as shown in Fig. 8, degree of financial leverage (DFL) and persistent EPS are also considered important because companies with high leverage and inconsistent EPS would be categorized as high-risk companies.

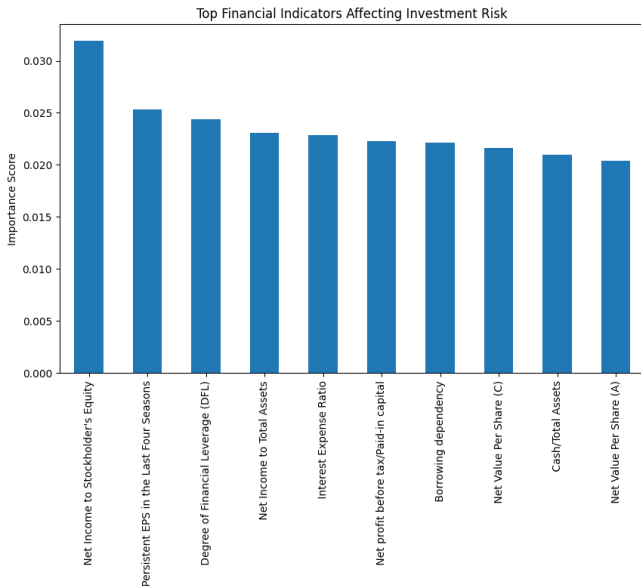


Fig. 8 Top Financial Indicators Affecting Investment Risk

Furthermore, ratios like cash/total assets, net worth/share, and interest expenses/ratio also add somewhat to the prediction task. This set of attributes offers additional information about liquidity and management of finance.

The resemblance in the results between Fig. 7 and Fig. 8 demonstrates the reliability and consistency of the model in choosing financial indicators.

G. Summary

From the above analyses, we observe that:

1. Random Forest yields the best prediction results.
2. Financial parameters are highly significant in predicting investment risks.

3. SHAP improves model interpretability.
4. Class imbalances pose significant difficulties.
5. Risk rating system increases practical utility.

In conclusion, the investment risk assessment system presented is efficient and incorporates both machine learning and explainable AI.

V. CONCLUSION

The paper developed a framework that leverages data-driven machine learning and explains AI to predict the investment risks based on different financial factors. The framework combines all stages of building, assessing, and deploying ML-based predictions, which makes it convenient to perform efficient and scalable risk evaluations.

First, we demonstrated that ensemble methods produce higher accuracy when predicting investment risks than Logistic Regression or Decision Trees. For example, Random Forest achieved impressive accuracy equal to 96.63%. Nevertheless, we observed that the class imbalance negatively impacted the model quality as it was evident from relatively low recall rate of high-risk companies. Thus, future work can be focused on overcoming the issue of class imbalance through resampling or cost-sensitive approaches.

We also found out that the features such as profitability ratio, leverage, and earnings consistency play an important role in determining the level of company risks. At the same time, the application of SHAP improved the model transparency, making it easy to understand how each feature affects the predicted investment risk level.

Moreover, risk score calculation allows using the probability of risks as a variable that ranges from 0 to 100, thus increasing the practical relevance of the solution.

Thus, our work showed that data-driven machine learning and explains AI could provide an efficient and accurate approach to predicting investment risks. Future work could include addressing the class imbalance problem and applying ML models for real-time data processing.

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